



Written Examination

23.08.23

Family Name: _____ First Name: _____

Program of Study: _____

Matr.-No.: _____

Information:

- Write your name and matriculation number on **every sheet of paper**.
- Make sure that all **20 pages** are present.
- Duration of the exam: **90 minutes**.
- **No additional aids** (notes, calculator, etc.) are allowed.
- Use a pen with **blue or black ink**. Text written with **pencils or red/green pens will NOT be graded**.
- Write **legibly**. Unreadable answers will not be graded.
- Answer each question on the provided sheet. If you need more space, use the two blank pages at the end or ask for more paper. In these cases, please make a short note in the answer box, so we can find your answer (e.g. "see last page").
- If you have to draw something to answer a question, multiple templates are provided. **Cross out wrong answers!**
- At the end of the exam you must hand in this cover sheet together with the question sheets and all additionally used paper.

With my signature I confirm that I have **read and understood** the information above.

Signature

Question:	1	2	3	4	5	6	Total	Grade
Points:	15	15	15	15	19	15	94	-
Score:								
Moderation:								

Question 1: Image Filtering and Pattern Recognition ($\Sigma = 15$)

- (a) How is **cross-correlation** filtering $G = F \otimes I$ defined, given an image I and a filter F ? Write the formula and explain the variables. (___/3)

- (b) We want to apply this 2D convolutional filter, shown below, as two separate 1D convolutional filters sequentially for more efficiency. Write two separate 1D convolutional filters that, applied sequentially, would produce the same result as the single 2D convolutional filter. (___/1)

-1	0	1
-2	0	2
-1	0	1

- (c) How many operations (addition, multiplication) would it take to apply sequentially two 1D 1×3 filters compared to one equivalent 2D filter 3×3 ? Give the number of operations per pixel for each case, in the general case ignoring boundary artifacts and assuming same padding at the image boundaries. (___/2)

- (d) How do you need to modify a 2D cross-correlation filter to represent a 2D convolution (___/1) filter?

- (e) Write down the steps of the Canny Edge Detector algorithm. (___/4)

- (f) Explain hysteresis thresholding. Why is it better than a simple thresholding? (___/2)

- (g) If we have a photo with salt and pepper noise, what problem(s) might we face when applying the Canny Edge Detection algorithm? What might be a possible approach to mitigate the problem? (___/2)

Question 2: Local Features ($\Sigma = 15$)

- (a) The Harris corner response function R can be expressed by the eigenvalues λ_1 and λ_2 (___/2) of the second-moment matrix. Explain the influence of the image content on these eigenvalues.

- (b) Briefly describe what automatic scale selection is in the context of feature descriptors (___/2) and how it works.

- (c) To which of these transformations is the Harris Detector invariant? Justify your answers!

i. Affine transformations?

(___/1)

ii. Euclidean transformations?

(___/1)

- (d) Once keypoints have been extracted, we want to extract features for matching using SIFT. What is the dimension of a SIFT feature descriptor? Explain how you get to this number in words.

- (e) Two images showing the same object from different viewpoints are given. We want to estimate a homography H from the left image to the right image (i.e. $\mathbf{p}_r = H\mathbf{p}_l$). Given are n point correspondences

$$\mathcal{P} = \{(\mathbf{p}_l^1, \mathbf{p}_r^1), (\mathbf{p}_l^2, \mathbf{p}_r^2), \dots, (\mathbf{p}_l^n, \mathbf{p}_r^n)\}$$

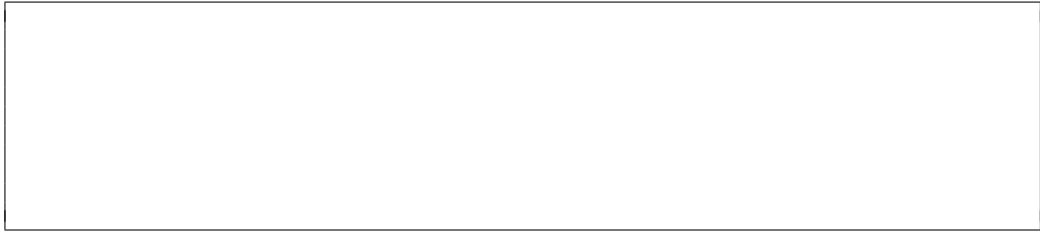
where $\mathbf{p}_l^i = (x_l^i, y_l^i, 1)^T$ is the i -th point in the left image in homogeneous coordinates. Analogously $\mathbf{p}_r^i$ is the point in the right image.

- i. To estimate H we solve an equation of the form $Ah = 0$ given by (___/6)

$$\begin{bmatrix} x_l & y_l & 1 & 0 & 0 & 0 & -x_r x_l & -x_r y_l & -x_r \\ 0 & 0 & 0 & x_l & y_l & 1 & -y_r x_l & -y_r y_l & -y_r \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \cdot \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = 0$$

Derive the equations for the matrix entries A given a single correspondence pair $(\mathbf{p}_l, \mathbf{p}_r)$.

- ii. How many correspondence pairs are needed to estimate a homography? Justify (___/1) your answer.



Question 3: Image Segmentation ($\Sigma = 15$)

(a) We want to segment an image using the Markov Random Fields formulation and the graph-cut algorithm. First we want to set up an energy function $E(\mathbf{x}, \mathbf{y})$:

- i. Write the energy function, explaining all the variables used. What is the role of (___/4) each of term?



- ii. Given the energy function $E(\mathbf{x}, \mathbf{y})$, and the goal of finding the segmentation label (___/1) $\hat{\mathbf{x}}$, what would be the optimization objective?



(b) Assume you have implemented a segmentation model using MRFs, where you can provide an initial rectangular region for the foreground and background regions.

- i. Assume that we express color likelihood by a histogram $p(\{r, g, b\}|c)$, expand the (___/1) posterior probability for the class label c given the observed pixel color $\{r, g, b\}$.



- ii. You used the graph-cut algorithm on top of MRFs to obtain segmentation results (___/1) from your model. However, you decided to apply your segmentation model *iteratively* reusing predicted masks as you did in the exercise, to get even better results. What parameters are updated every iteration and why?

- (c) Assume you are given multiple point clouds, such as in a Figure 1, where each point cloud is a set of points in a 3D space (XYZ), where the majority of the points lie in a horizontal plane (road). For each point cloud, we want to find a plane that fits the majority of the points (find the road plane). However, we have constraints: we do not have labels, and we want to use only positional information. We decided to use RANSAC for this task.

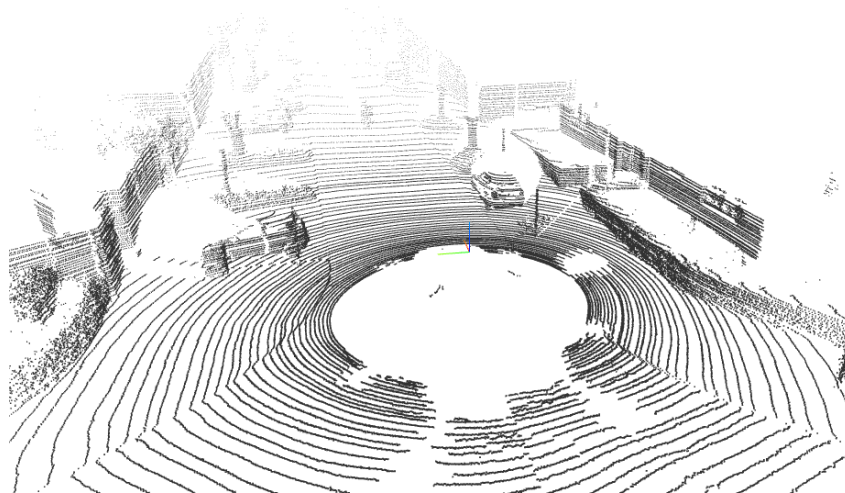


Figure 1: Point Cloud obtained from a LiDAR Scan installed on a car.

- i. Describe the steps of the RANSAC algorithm for the simple case of line fitting. (___/4)

- ii. What changes do we need to make to RANSAC to apply it to the task of ground plane fitting in 3D point clouds? (___/1)

- iii. Given the same constraints as before, explain why the following other algorithms presented in the lecture would be a poor choice for finding the ground plane in a 3D point cloud:

1) K-Means

(___/1)

2) MaxFlow

(___/1)

3) Mixture of Gaussians

(___/1)

Question 4: Sliding-Window Object Detection..... ($\Sigma = 15$)

- (a) Briefly outline the overall approach of sliding-window object detection. (___/2)

- (b) Explain the concept of a linear SVM classifier, i.e. what do we aim to find, what do we aim to maximize? Use words, not formulas. (___/2)

- (c) What is the kernel trick, how does it work and why is it useful? (___/2)

(d)

(____/2)



Given a dataset of 1000 RGB images of pedestrians, as shown above. Is it possible to train a linear SVM with HOG features on the given dataset to detect pedestrians in a sliding-window fashion? Justify your answer.

- (e) Boosting is used to build a strong classifier out of several weak classifiers. After training a new weak classifier $h_m(x)$, what values do we need to adjust and in which direction? (____/2)

- (f) Explain how the weak classifiers used in the Viola-Jones detector work, i.e. how are the features computed and classified? (____/2)

- (g) The Roerei-detector extends the Viola-Jones detector by using an irregular pattern (___/3) of rectangles to compute the features for detection. Assume we want to use different shapes instead of rectangles, e.g. circles or triangles. Explain the most significant disadvantage of this idea. What could be a potential advantage?

Question 5: Convolutional Neural Networks ($\Sigma = 19$)

- (a) What is the main difference and advantage of a CNN over the traditional fully-connected Multi-Layer Perceptron (MLP)? (___/1)

- (b) We define a block in the model that consists of three consecutive layers:

1. Convolution
2. ReLU
3. BatchNorm

Suppose you want to extract feature maps from RGBD (D is an additional input channel for depth) images of the size 100×100 using a **single** block defined above. We use a kernel size of 3×3 , with a bias, stride of 1, and zero padding of the size of 1.

- i. How many learned parameters do we need for a single block if the output is the depth of 10? (___/1)

- ii. What is the number of parameters if we use an image of size 1000×1000 ? (___/1)

- iii. If we were to stack three more identical blocks, what is the receptive field that activates a pixel in the 4th layer of the CNN? Ignore the effect of BatchNorm on the receptive field. (___/1)

- iv. You want to keep the same number of convolutions, but improve the receptive field (___/1) of the model by adding some additional layers to the model. What methods can you suggest to improve the receptive field?

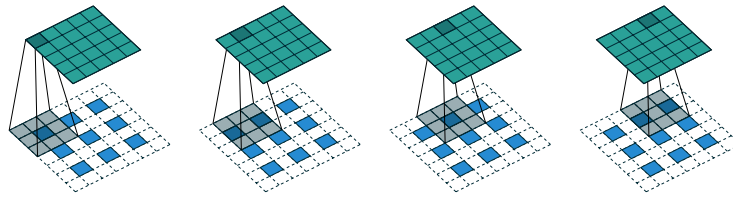
- (c) What would the 4×4 input in the figure below look like after sequentially applying (___/3) 2×2 Max Pooling, and Max Unpooling operations with stride $s = 2$? What is the difference between Unpooling and Max Unpooling?

3	1	1	8
6	2	2	4
2	5	2	3
1	3	1	7

- (d) For each set of images, write the name of the convolution operation, the kernel size k , the input size i , the stride s , and the padding size p .

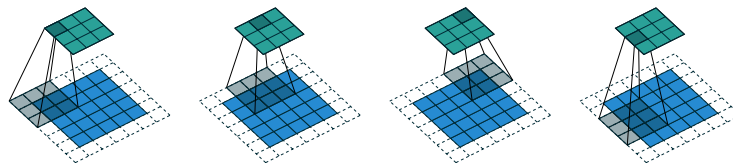
i.

(___/2)



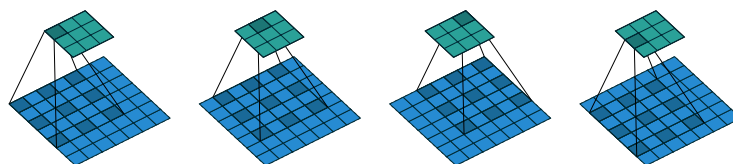
ii.

(___/2)



iii.

(___/2)



- (e) Given a minibatch of n samples (x_i, t_i) , write a formula for the Cross-Entropy loss and (___/2) explain the variables used.

- (f) Is it possible to use cross-entropy loss for both semantic segmentation and classification? (___/1) If not, suggest an alternative.

- (g) Why do we want to use an encoder-decoder architecture for segmentation? Why do (___/2) we want to use skip connections between encoder and decoder?

Question 6: 3D Reconstruction ($\Sigma = 15$)

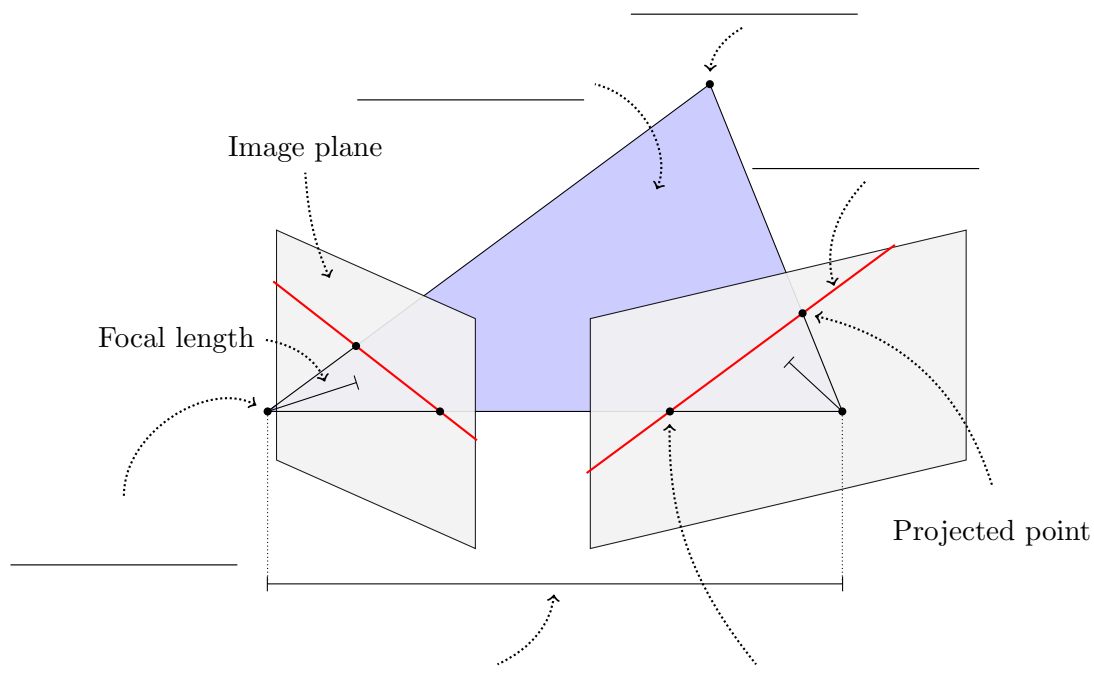
- (a) Write a function that computes the camera center based on the camera projection (___/2) matrix. Use python syntax or pseudo-code and assume numpy is imported. Loops are not allowed in this task. Small syntax errors are allowed as long as the meaning is clear.

```

1 import numpy as np
2
3 def camera_center_from_projection_matrix(P):
4     """
5     Args:
6         P: projection matrix
7
8     Returns:
9         center: camera center
10    """
11
12    # START HERE
13
14
15
16
17
18
19
20
21
22
23
24    return center

```

- (b) Given is a stereo setup with two cameras. Name all the requested elements in the (___/3) following image.



- (c) Derive the epipolar constraint, given the calibration (rotation $\mathbf{R}$ and translation $\mathbf{T}$) (___/6) and the points $\mathbf{X}'$ and $\mathbf{X}$. Explicitly give the equation for the essential matrix $\mathbf{E}$.

- (d) Assume you want to estimate a homography with RANSAC and you know that the proportion of outliers is given by δ . Derive the equation for the minimum number of iterations k needed such that the probability $P(\text{fail})$ of all iterations to fail is less than ε . (___/2)

Hints: $\log(a^b) = b \log(a)$ and do not forget to flip inequality signs if you divide by a negative number.

- (e) In the following we want to consider projective Structure-from-Motion (SfM), where we (____/2) are given m images of n fixed 3D points $\mathbf{X}_j$ that are visible in each image. Projective SfM can only be solved when

$$2mn \geq 11m + 3n - 15$$

Explain the reasoning behind the different terms.

Name:

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Use these pages if you run out of space in the solution box.
Please add a reference pointing to this page from the box of the question.

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